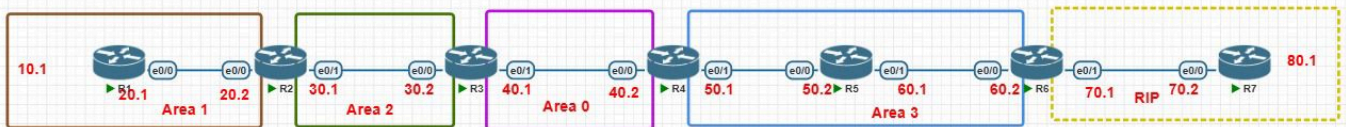


## OSPF AREAWISE



```
R1
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:36    20.0.0.2     Ethernet0/0
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Loopback0
L       10.0.0.1/32 is directly connected, Loopback0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.1/32 is directly connected, Ethernet0/0
R1(config)#
```

```
R2
O IA 50.0.0.0/8 [110/30] via 30.0.0.2, 00:20:59, Ethernet0/1
O IA 60.0.0.0/8 [110/40] via 30.0.0.2, 00:16:23, Ethernet0/1
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
1.1.1.1          1     FULL/DR         00:00:38    20.0.0.1     Ethernet0/0
3.3.3.3          1     FULL/BDR        00:00:37    30.0.0.2     Ethernet0/1
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/32 is subnetted, 1 subnets
O       10.0.0.1 [110/11] via 20.0.0.1, 01:37:48, Ethernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.2/32 is directly connected, Ethernet0/0
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.1/32 is directly connected, Ethernet0/1
O IA 40.0.0.0/8 [110/20] via 30.0.0.2, 01:19:00, Ethernet0/1
O IA 50.0.0.0/8 [110/30] via 30.0.0.2, 00:24:13, Ethernet0/1
O IA 60.0.0.0/8 [110/40] via 30.0.0.2, 00:19:37, Ethernet0/1
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#
```



```
R3
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
4.4.4.4          1     FULL/BDR        00:00:32    40.0.0.2     Ethernet0/1
2.2.2.2          1     FULL/DR         00:00:32    30.0.0.1     Ethernet0/0
R3(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/0
L       30.0.0.2/32 is directly connected, Ethernet0/0
    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/8 is directly connected, Ethernet0/1
L       40.0.0.1/32 is directly connected, Ethernet0/1
O IA   50.0.0.0/8 [110/20] via 40.0.0.2, 00:25:15, Ethernet0/1
O IA   60.0.0.0/8 [110/30] via 40.0.0.2, 00:20:39, Ethernet0/1
R3(config)#
R3(config)#
R3(config)#
R3(config)#
```



R4

```
Router#
Router#
Router#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
3.3.3.3          1     FULL/DR         00:00:37    40.0.0.1     Ethernet0/0
5.5.5.5          1     FULL/BDR        00:00:33    50.0.0.2     Ethernet0/1
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  30.0.0.0/8 [110/20] via 40.0.0.1, 00:26:50, Ethernet0/0
      40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      40.0.0.0/8 is directly connected, Ethernet0/0
L      40.0.0.2/32 is directly connected, Ethernet0/0
      50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      50.0.0.0/8 is directly connected, Ethernet0/1
L      50.0.0.1/32 is directly connected, Ethernet0/1
O      60.0.0.0/8 [110/20] via 50.0.0.2, 00:22:08, Ethernet0/1
Router#
```

```
R5
Router>
Router>
Router>
Router>
Router>en
Router#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
6.6.6.6          1     FULL/BDR        00:00:37    60.0.0.2     Ethernet0/1
4.4.4.4          1     FULL/DR         00:00:32    50.0.0.1     Ethernet0/0
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  30.0.0.0/8 [110/30] via 50.0.0.1, 00:23:26, Ethernet0/0
O IA  40.0.0.0/8 [110/20] via 50.0.0.1, 00:23:26, Ethernet0/0
      50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      50.0.0.0/8 is directly connected, Ethernet0/0
L      50.0.0.2/32 is directly connected, Ethernet0/0
      60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      60.0.0.0/8 is directly connected, Ethernet0/1
L      60.0.0.1/32 is directly connected, Ethernet0/1
Router#
Router#
Router#
```



```
R6
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#en
% Incomplete command.

Router(config)#emd
^
% Invalid input detected at '^' marker.

Router(config)#end
Router#sh
*Aug 22 16:24:55.134: %SYS-5-CONFIG_I: Configured from console by console
Router#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
5.5.5.5          1     FULL/DR         00:00:31    60.0.0.1     Ethernet0/0
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

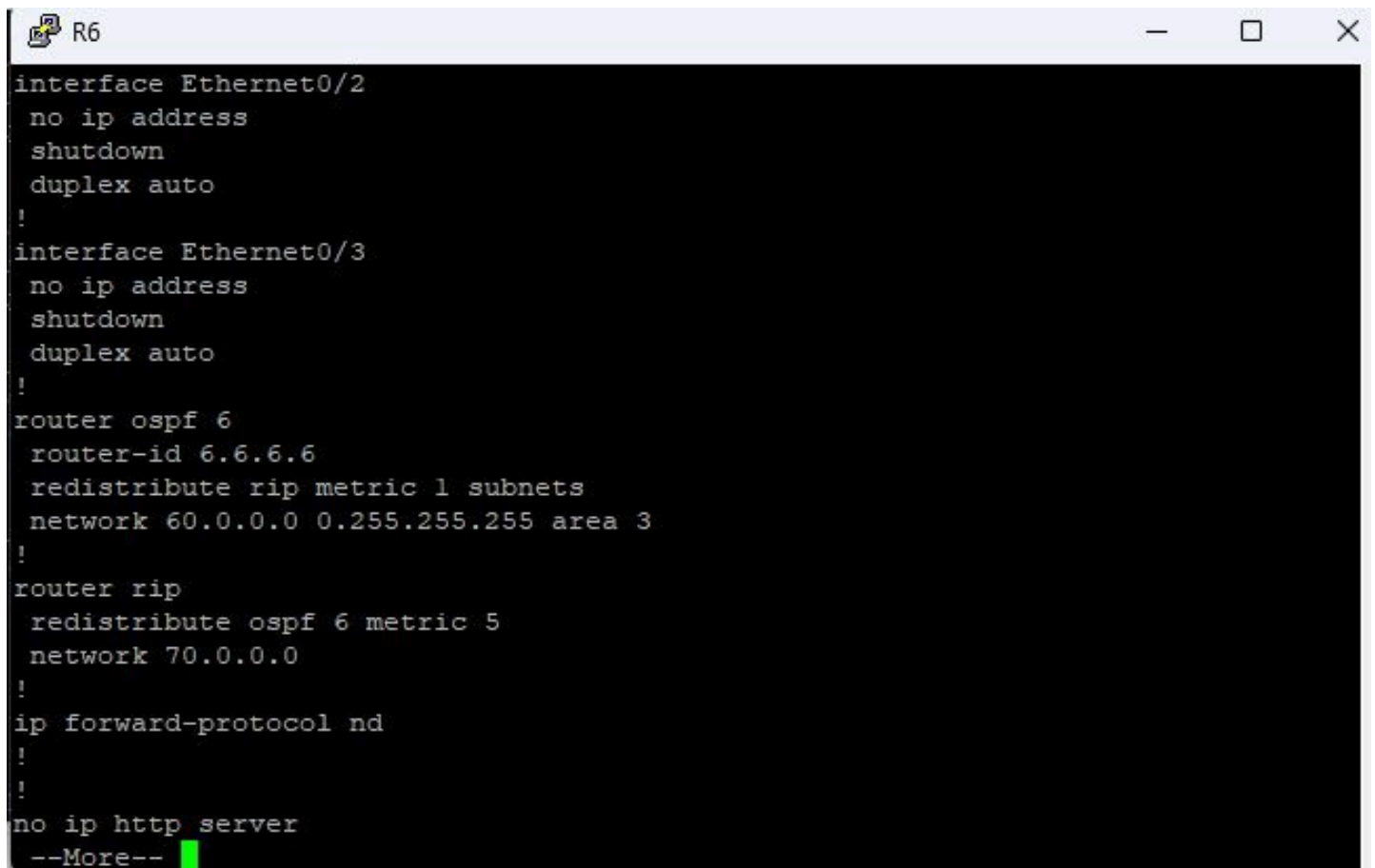
O IA  30.0.0.0/8 [110/40] via 60.0.0.1, 00:11:25, Ethernet0/0
O IA  40.0.0.0/8 [110/30] via 60.0.0.1, 00:11:25, Ethernet0/0
O     50.0.0.0/8 [110/20] via 60.0.0.1, 00:11:25, Ethernet0/0
      60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     60.0.0.0/8 is directly connected, Ethernet0/0
L     60.0.0.2/32 is directly connected, Ethernet0/0
      70.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     70.0.0.0/8 is directly connected, Ethernet0/1
L     70.0.0.1/32 is directly connected, Ethernet0/1
Router#
```

```
R7#
R7#
R7#
R7#
R7#
R7#
R7#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    70.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       70.0.0.0/8 is directly connected, Ethernet0/0
L       70.0.0.2/32 is directly connected, Ethernet0/0
    80.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       80.0.0.0/8 is directly connected, Loopback1
L       80.0.0.1/32 is directly connected, Loopback1
R7#
```



A screenshot of a Cisco IOS configuration window titled 'R6'. The window has a standard macOS-style title bar with a close button (X) on the right. The background is black, and the text is white. The configuration commands are as follows:

```
interface Ethernet0/2
  no ip address
  shutdown
  duplex auto
!
interface Ethernet0/3
  no ip address
  shutdown
  duplex auto
!
router ospf 6
  router-id 6.6.6.6
  redistribute rip metric 1 subnets
  network 60.0.0.0 0.255.255.255 area 3
!
router rip
  redistribute ospf 6 metric 5
  network 70.0.0.0
!
ip forward-protocol nd
!
!
no ip http server
--More--
```

A green cursor is visible at the end of the '--More--' line.

```
R3
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/0
L       30.0.0.2/32 is directly connected, Ethernet0/0
  40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/8 is directly connected, Ethernet0/1
L       40.0.0.1/32 is directly connected, Ethernet0/1
O IA   50.0.0.0/8 [110/20] via 40.0.0.2, 01:03:53, Ethernet0/1
O IA   60.0.0.0/8 [110/30] via 40.0.0.2, 00:59:17, Ethernet0/1
O E2   70.0.0.0/8 [110/1] via 40.0.0.2, 00:00:29, Ethernet0/1
O E2   80.0.0.0/8 [110/1] via 40.0.0.2, 00:00:29, Ethernet0/1
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
```

```
R6
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA 30.0.0.0/8 [110/40] via 60.0.0.1, 00:04:06, Ethernet0/0
O IA 40.0.0.0/8 [110/30] via 60.0.0.1, 00:04:06, Ethernet0/0
O    50.0.0.0/8 [110/20] via 60.0.0.1, 00:04:06, Ethernet0/0
    60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    60.0.0.0/8 is directly connected, Ethernet0/0
L    60.0.0.2/32 is directly connected, Ethernet0/0
    70.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    70.0.0.0/8 is directly connected, Ethernet0/1
L    70.0.0.1/32 is directly connected, Ethernet0/1
R    80.0.0.0/8 [120/1] via 70.0.0.2, 00:00:25, Ethernet0/1
Router(config)#
Router(config)#
Router(config)#
```



```
R2
interface Ethernet0/1
 ip address 30.0.0.1 255.0.0.0
 duplex auto
!
interface Ethernet0/2
 no ip address
 duplex auto
!
interface Ethernet0/3
 no ip address
 duplex auto
!
router ospf 2
 router-id 2.2.2.2
 area 2 virtual-link 3.3.3.3
 network 20.0.0.0 0.255.255.255 area 1
 network 30.0.0.0 0.255.255.255 area 2
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
--More--
```

```
R3
duplex auto
!
interface Ethernet0/2
 no ip address
 duplex auto
!
interface Ethernet0/3
 no ip address
 duplex auto
!
router ospf 3
 router-id 3.3.3.3
 area 2 virtual-link 2.2.2.2
 network 30.0.0.0 0.255.255.255 area 2
 network 40.0.0.0 0.255.255.255 area 0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
--More--
```

```
R1
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Loopback0
L       10.0.0.1/32 is directly connected, Loopback0
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.1/32 is directly connected, Ethernet0/0
O IA   30.0.0.0/8 [110/20] via 20.0.0.2, 00:02:32, Ethernet0/0
O IA   40.0.0.0/8 [110/30] via 20.0.0.2, 00:00:59, Ethernet0/0
O IA   50.0.0.0/8 [110/40] via 20.0.0.2, 00:00:59, Ethernet0/0
O IA   60.0.0.0/8 [110/50] via 20.0.0.2, 00:00:59, Ethernet0/0
O E2   70.0.0.0/8 [110/1] via 20.0.0.2, 00:00:54, Ethernet0/0
O E2   80.0.0.0/8 [110/1] via 20.0.0.2, 00:00:54, Ethernet0/0
R1(config)#
R1(config)#
R1(config)#
```

```
Router(config)#
Router(config)#
Router(config)#
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

R      10.0.0.0/8 [120/5] via 70.0.0.1, 00:00:23, Ethernet0/0
R      20.0.0.0/8 [120/5] via 70.0.0.1, 00:00:23, Ethernet0/0
R      30.0.0.0/8 [120/5] via 70.0.0.1, 00:00:23, Ethernet0/0
R      40.0.0.0/8 [120/5] via 70.0.0.1, 00:00:23, Ethernet0/0
R      50.0.0.0/8 [120/5] via 70.0.0.1, 00:00:23, Ethernet0/0
R      60.0.0.0/8 [120/5] via 70.0.0.1, 00:00:23, Ethernet0/0
       70.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      70.0.0.0/8 is directly connected, Ethernet0/0
L      70.0.0.2/32 is directly connected, Ethernet0/0
       80.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      80.0.0.0/8 is directly connected, Loopback2
L      80.0.0.1/32 is directly connected, Loopback2
Router(config)#do ping 10.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router(config)#
```